

Brian Woods

Applied Data Scientist · Causal Inference & Production ML · Marketplaces, Credit & Growth
San Francisco, CA · me@brian.biz · linkedin.com/in/brian-r-woods · brian.biz

SUMMARY

Data scientist and engineer (M.S., UC Berkeley, 2026) whose published work is exactly the statistical toolkit this role names: causal inference on messy observational data, with regression discontinuity, fixed-effects panels, placebo tests, and Bayesian models, defended live under technical scrutiny. I productionize what I model: my live trading platform serves gradient-boosted and reinforcement-learning models under drift monitoring, champion/challenger evaluation, and staged rollout, all built by hand. Before data science I spent a career on the problems this team owns: credit mechanics at two fintech companies, and growth engineering at one of the largest social networks of its era. My recent domain is crypto-market data; wholesale retail would be new, and carrying rigorous methods into new domains is what the 30 years have mostly consisted of.

SELECTED WORK

Published Causal Research · 2026 (brian.biz/optimum)

- Two published studies of block-propagation latency economics on Ethereum and Solana mainnet: regression discontinuity at a protocol deadline, null-calibrated difference-in-differences, CUPAC variance reduction, PyMC tail modeling, and a per-operator revenue model, on data from three self-operated collector nodes with an automated pipeline that re-estimates and republishes daily. Methods defended live in technical interviews.

Production ML Platform on Solana (sole owner-operator) · 2023-Present

- Real-time serving that is both performant and explainable: gradient-boosted models (LightGBM/XGBoost) and RL policies (PPO, CQL, behavioral cloning; PyTorch) over gRPC streaming ingest at 15M+ transactions per run, with feature pipelines, drift monitoring, champion/challenger evaluation, staged shadow rollout, and fail-closed gating. Trading P&L to date is flat to negative and reported publicly; the measurement framework that makes that number trustworthy is the discipline I bring.

Growth Engineering at Scale · Bebo, 2005-2009

- Ran growth engineering at one of the world's largest social networks at its peak (~250K registrations/day): funnel measurement, cohort analysis, and experiment-driven acquisition and retention across international markets. The LTV-and-lifecycle problem this role describes, at consumer-internet scale.

VitalFoods.ai, ML Recommendation Platform (UC Berkeley capstone, team project; live)

- Designed the original RotatE knowledge-graph embedding models on USDA FoodData Central data and built the full production website; a teammate led embedding training. Ran all SageMaker and EC2 infrastructure. Live with real users.

EXPERIENCE

Independent Engineer & Data Scientist, Trading Systems & Published Research (self-employed) · 2023-Present

The systems above, built concurrently with the UC Berkeley M.S. Started with one collaborator who has since moved on.

Chief Technology Officer, AGEN.AI (bootstrapped, never raised funding) · 2023-2024

Led architecture and delivery of the company's AI product across a small team.

Senior Software Developer, Umee (company wound down) · 2022

Built the credit mechanics of a blockchain lending platform in Go: collateralization, borrow limits, and interest accrual as on-chain state, where an underwriting error is an immediate loss.

Senior Protocol Engineer, Sifchain (company wound down) · 2021-2022

Shipped financial products in Go and Python on the Cosmos SDK; hardened inter-blockchain communication (IBC) in production.

Core Developer, Augur Project · 2017-2020

Built decentralized prediction-market infrastructure (TypeScript/JavaScript): a two-sided marketplace where prices aggregate information.

Core Backend Developer, Dispatch Labs · 2015-2017

Owned the EVM implementation and its correctness.

Director of Engineering, HelloBit · 2013-2015

Owned architecture, hiring, and delivery for a Bitcoin remittance startup moving real money across borders, with the fraud and risk exposure that implies.

Director of Engineering, AnythingSocial · 2010-2013

Led delivery of mobile and Rails consumer applications across distributed teams.

Software Engineer, Tickle · 2001-2005

Built a content-discovery product driven by user behavior data.

Senior Engineer, Yahoo Mail, Yahoo Inc. · 1996-2001

Wrote the C++ mail transport agent handling 6B+ messages/month and built anti-spam classification against an adversarial workload (6 patents granted).

EDUCATION

M.S., Information & Data Science (AI concentration), UC Berkeley · 2024-2026

AI concentration. Capstone shipped as VitalFoods.ai (above). Completed while building and operating the trading platform.

B.S., Computer Science & Economics, University of Colorado Boulder · 1996**PATENTS & CERTIFICATIONS**

- 6 U.S. patents in bulk-email processing and anti-spam (Yahoo Mail).
- Cryptography I (Stanford) · GCP Big Data & ML Fundamentals.

SKILLS

Causal inference & experimentation: regression discontinuity, difference-in-differences, fixed-effects panels with clustered inference, placebo and calibration tests, CUPAC variance reduction, Bayesian modeling (PyMC), bootstrap uncertainty, experiment design and pre-registration

Machine learning: LightGBM/XGBoost, PyTorch, reinforcement learning (PPO, CQL, behavioral cloning), feature engineering and pipelines, drift monitoring, champion/challenger evaluation, explainable model design for decisions with money on the line

Engineering: Python (pandas, PyMC), SQL, Rust (extension modules in the Python interpreter), Go, C++, TypeScript/JavaScript; real-time streaming (gRPC); AWS (EC2, SageMaker) run hands-on; production MLOps built from scratch

Domain: credit and lending mechanics (two fintech companies), two-sided marketplaces, growth/funnel/cohort analytics at consumer scale, adversarial classification (anti-spam, anti-fraud adjacent)